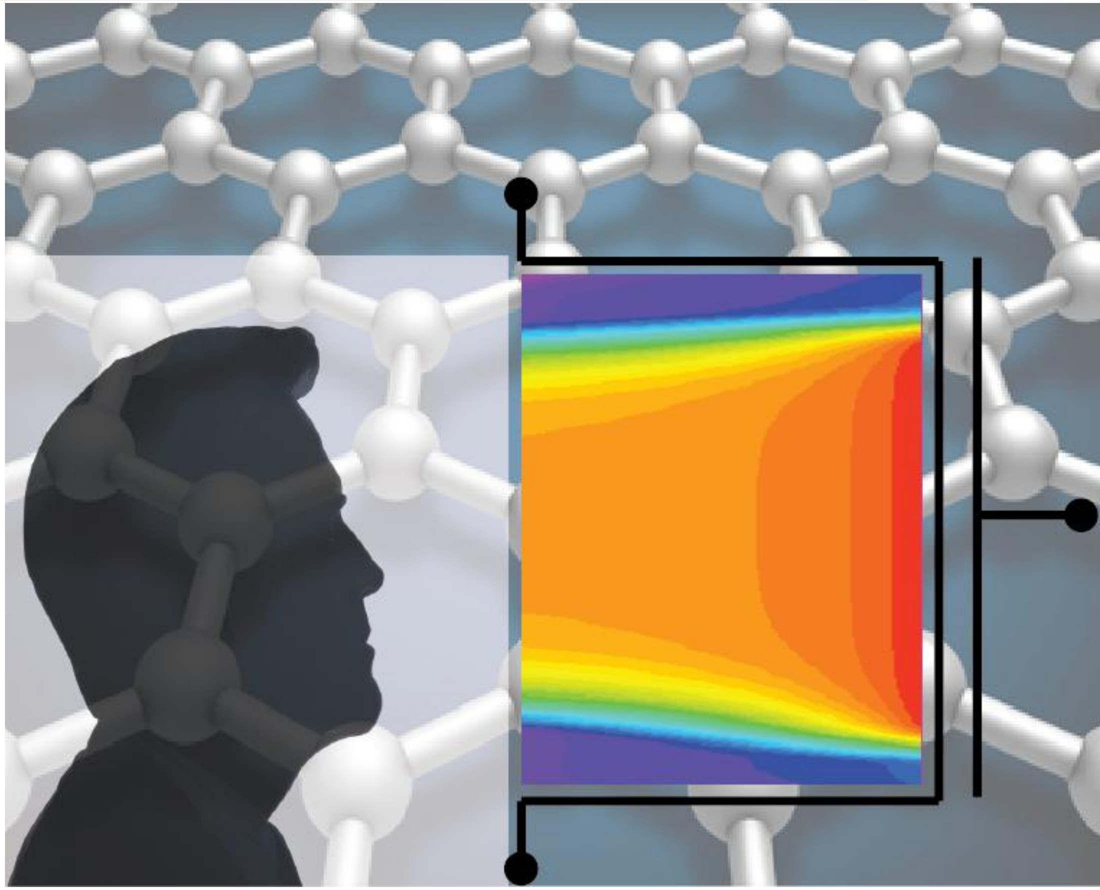
ADVANCED SOLID-STATE DEVICES

Week-14, Lecture-2



Sneh Saurabh
24th April, 2025

Ballistic and Diffusive Transport

What is Ballistic Transport?

What is typical physical gate length of transistors in 7 nm technology nodes?

- Around 20 nm

What will happen when gate length goes below 10 nm (say 5 nm)?

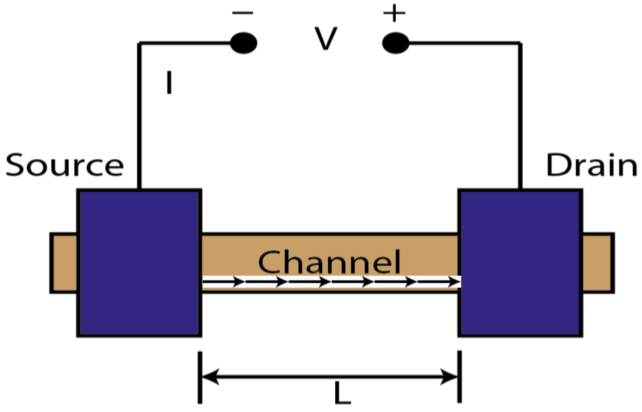
- Mean free path length of electrons in Silicon is 10-20 nm
- Electron may not encounter any scattering in the channel (transport becomes **ballistic**)

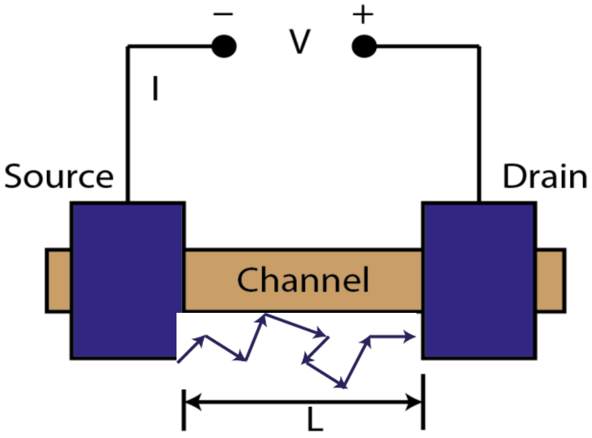
Ballistic Transport (like a bullet):

- Occurs when the channel length of the transistor is shorter than the mean free path of the electrons
- Carriers travel without or minimal scattering

Diffusive Transport (with scattering)

- Channel contains numerous scatterers

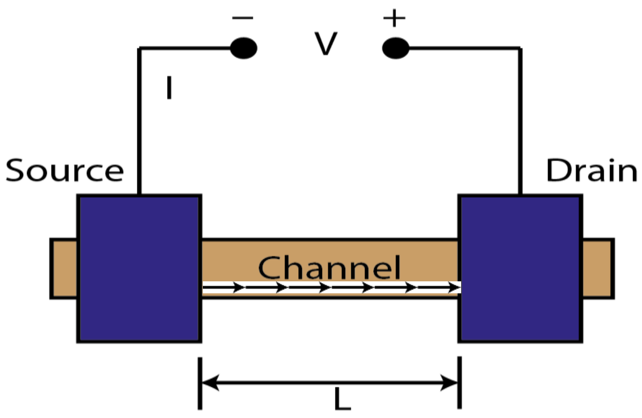




What happens in Ballistic Transport?

Behavior of ballistic transport

- Current is not limited by collisions (unlike diffusive transport)
- Drive current and performance can be improved
- Limited by how efficiently carriers are injected and collected at the source and drain
  - Contact resistance becomes dominant
  - There are physics-imposed limits on how good a contact we can make



When is Ballistic Transport important?

- At advanced process nodes (~20 nm gate length): a mix of ballistic and diffusive transport
- With gate length reduction: ballistic transport becomes significant
- Some devices can have dominant ballistic transport: CNT FETs (CNTFETs) and III-V High Electron Mobility Transistor (HEMT)

Choice of Channel Materials

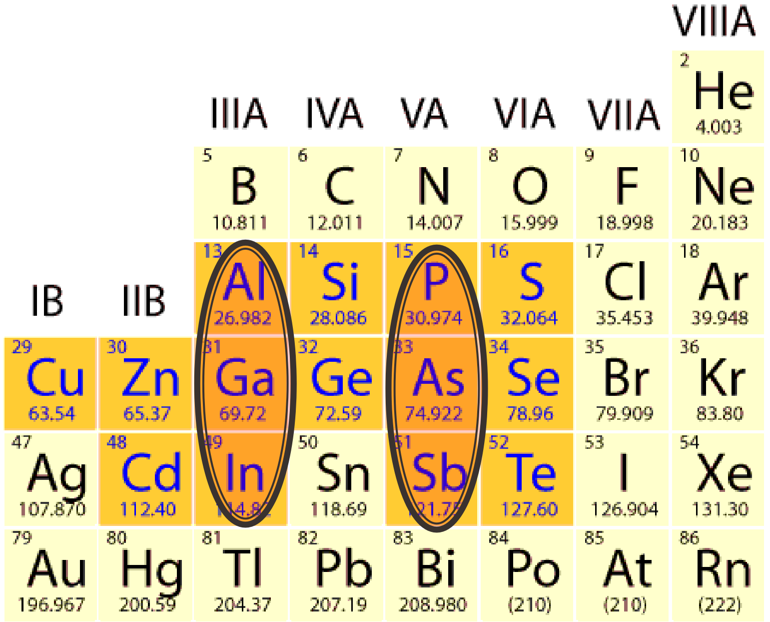
Channel Material

III-V Semiconductors

III-V Semiconductors: Basics

What are III-V Semiconductors?

- III = elements of group IIIA in the periodic table
  - ➢ Al, Ga, In
- V = elements of group VA in the periodic table
  - ➢ P, As, Sb



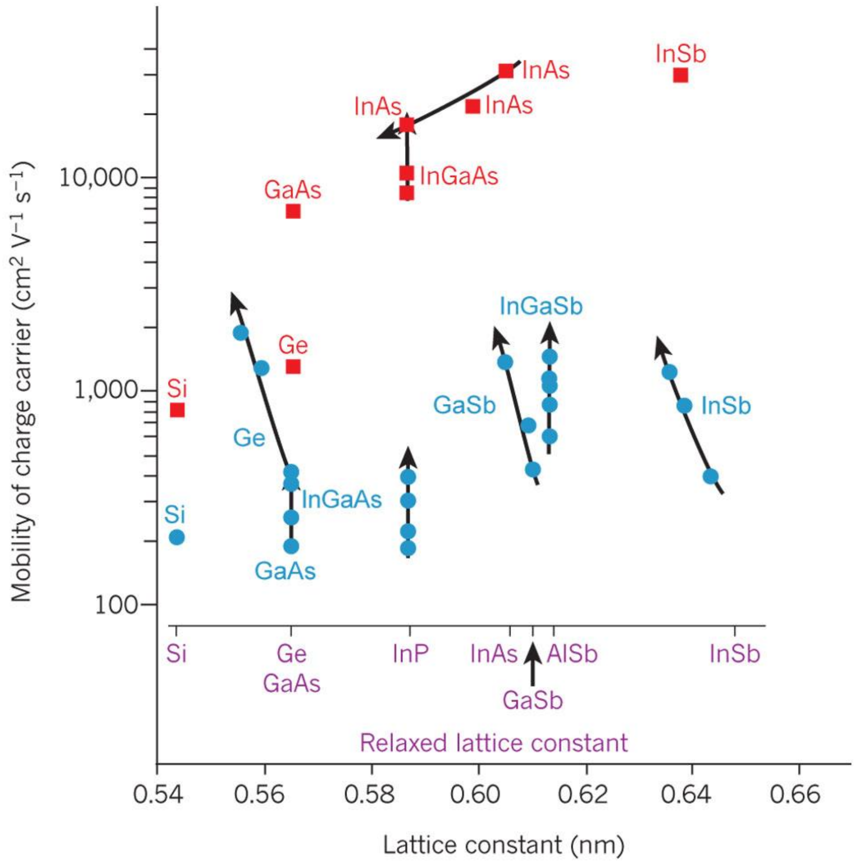
- <u>Binary Compounds</u>: $A^{III}Z^{V}$
  - ➢ Example: GaAs, InP, InAs etc.
- <u>Ternary Compounds</u>: combine two binary compounds $A^{III}Z^{V}$ and $B^{III}Z^{V}$ in the ratio $x{:}(1-x)$ to obtain ternary $A_x^{III}B_{(1-x)}^{III}Z^{V}$
  - ➢ Example: $In_{0.53}Ga_{0.47}As$, $In_{0.05}Ga_{0.95}Sb$ etc.

III-V Semiconductors: Properties

- Mobility of carriers in III-V semiconductors is appreciably higher compared to the mobility of carriers in silicon

- For example:
  - ➢ $Mobility\ of\ electrons\ in\ InAs \leq 40000 cm^2/Vs$
  - ➢ $Mobility\ of\ electrons\ in\ silicon \leq 1400 cm^2/Vs$

- Excellent mobility of III-V semiconductors can be utilized in transistors to obtain high drain current
  - ➢ $I_D = \frac{\kappa \epsilon_0}{t_{ox}} \mu \left(\frac{W}{L}\right) \frac{(V_{GS}-V_T)^2}{2}]$
  - ➢ Speed/Performance of devices improve



Source: http://www.nature.com/nature/journal/v479/n7373/fig_tab/nature10677_F1.html

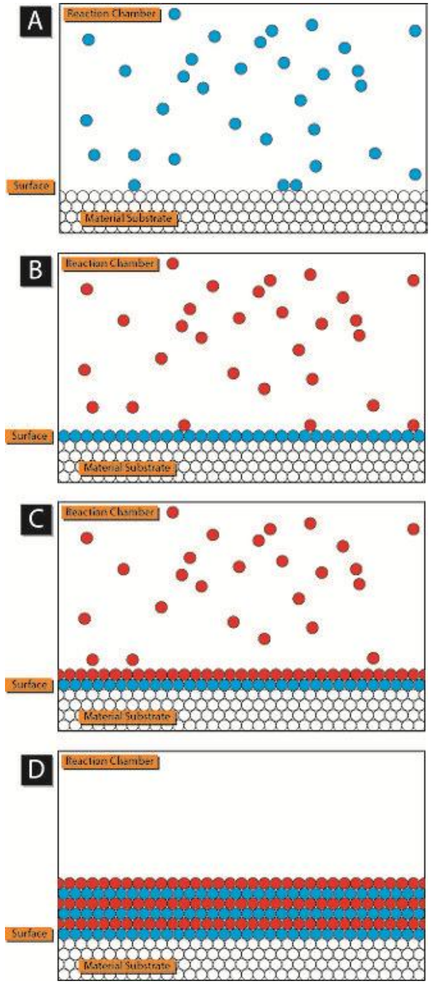
III-V Semiconductors: Challenges and Solution

Challenges

- MOSFETs realized using III-V semiconductors were of bad quality until recently
- III-V semiconductors do not have native oxide that have *good interface quality* [interface between III-V semiconductor and its oxide]
- Due to high density of interface states at III-V semiconductor/dielectric interface, the effect of gate potential on *modulating potential energy in the channel* was diminished

Recent Solution

- *Atomic Layer Deposition (ALD)*
- Atomic layer precision achieved
- “Self Cleaning Effect”
- High quality III-V semiconductor/dielectric interface can be obtained



Source: Wikipedia

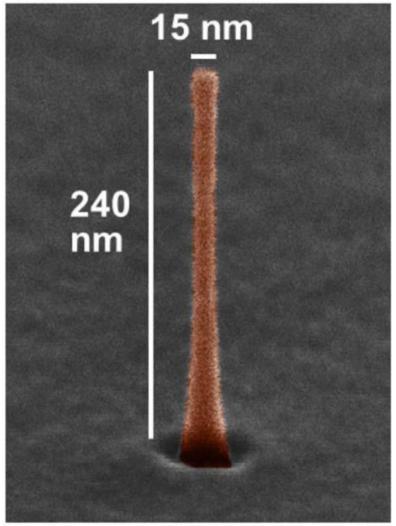
III-V Semiconductors: Future CMOS

- Employ III-V semiconductor materials as channel
    - ➢ Exploit high mobility of III-V semiconductors

- Most promising III-V semiconductor for future CMOS
    - ➢ n-channel MOSFETs: InGaAs
    - ➢ p-channel MOSFETs: InGaSb
- In future III-V semiconductor based transistors are highly possible

- III-V semiconductor based MOSFET being explored aggressively in different architectures
    - ➢ Planer
    - ➢ FinFET
    - ➢ Nanowire

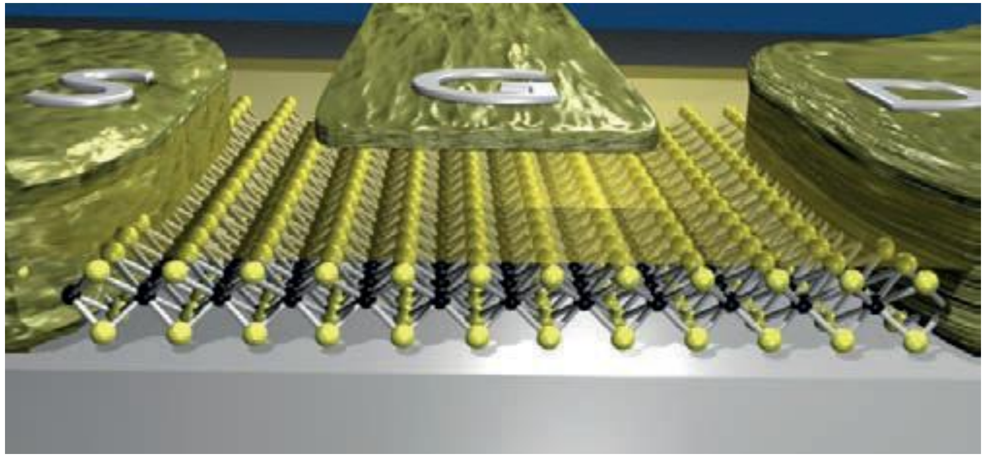
InGaAs nanowire fabricated



Source: Del Alamo, Jesús A., et al. "Nanometer-Scale III-V MOSFETs." *IEEE Journal of the Electron Devices Society* 4.5 (2016): 205-214.

2-D Devices

2-D materials: Motivation



- Widely explored 2-D materials for transistors:
  - Graphene
  - Transition Metal Dichalcogenides [TMDs]

- We have seen that thin channel helps in tackling short channel effects and improving characteristics of the transistors

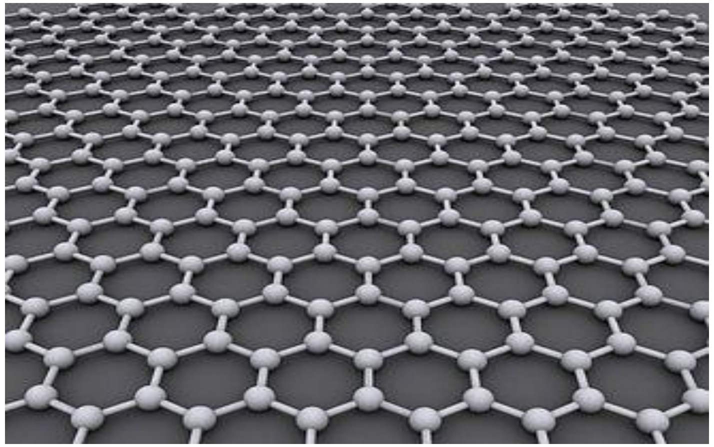
- Transistors realized using 2D materials represent the ***ultimate thin channel***
  - Improves gate control over the channel potential
  - Decreases short-channel effects
  - Provides atomic level control of thickness

2-D Devices

Graphene-based FETs

Graphene: Basics

- Graphene is a two-dimensional (2-D) *atomic layer of carbon* atoms that is sufficiently *isolated from the environment* to be considered free-standing



- The carbon atoms are arranged in a hexagonal lattice

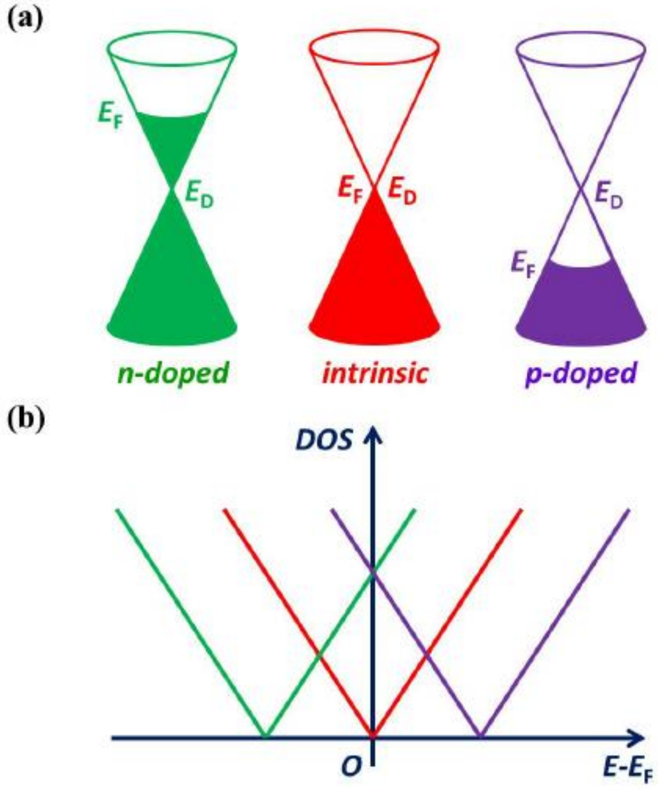
- In 2004 Geim and Novoselov extracted single-atom-thick crystallites from bulk graphite

- *Scotch tape technique*: pulled graphene layers from graphite and transferred them onto thin silicon dioxide (SiO_2) on a silicon wafer

Graphene: Properties

- It has a very high carrier mobility at room temperature
- High thermal conductivity
- Very high tensile strength.

- The band structure of graphene has a linear dispersion relationship with the momentum proportional to the energy
- In a large-area graphene, the conduction band and the valence bands in graphene are conical and meet at a point
    - $\Rightarrow$ Zero bandgap
    - $\Rightarrow$ Metallic Properties



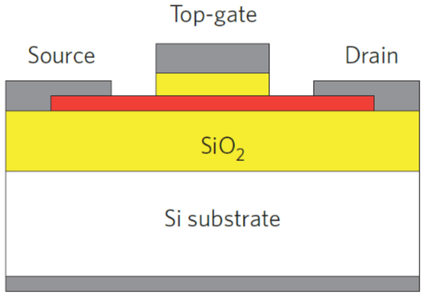
Source: Gao, Li. "Probing Electronic Properties of Graphene on the Atomic Scale by Scanning Tunneling Microscopy and Spectroscopy." *Graphene and 2D Materials* 1.1 (2014)

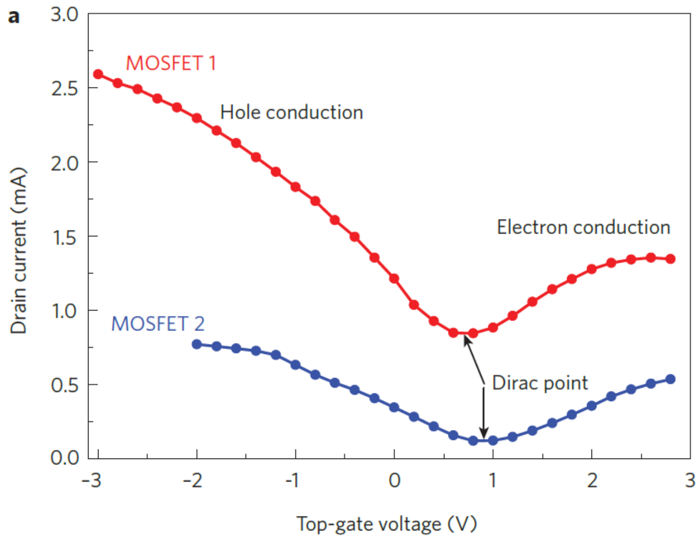
Graphene: Transistor

- Graphene FETs can be realized in many different ways

- Very good control of gate
- Immunity against short-channel effects
- Can be scaled to very small dimensions

- Zero bandgap does not allow to effectively turn the transistor OFF
- ON-current/OFF-current ratio typically 2-20 for Graphene-based FETs
  - ➢ Not acceptable for logic circuits: required **greater than 10^4**





Source: F. Schwierz, "Graphene Transistors", *Nature Nanotechnology*, pp. 487-496, May 2010

Graphene: Prospects

- Creating a bandgap in graphene in a controlled manner is THE BIGGEST challenge for graphene
- Rapid progress is being made
- Graphene interconnects are also promising for future application

2-D Devices

TMD-based FETs

TMD: Basics

- TMD: Transition Metal Dichalcogenides

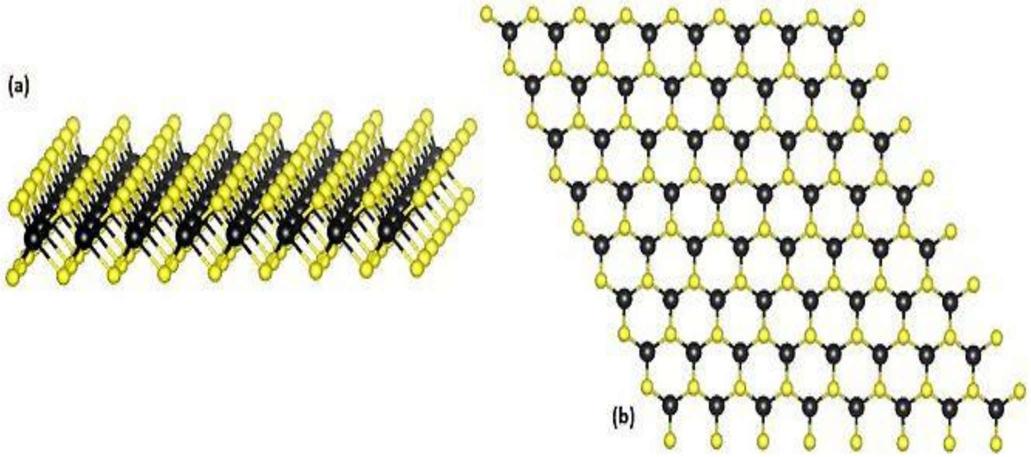
- General Formula: MX_2
  - ➢ $M \equiv Transition\ Metal$ [Example: Mo, W, Ti, Zr, Ta, Nb]
  - ➢ $X \equiv Chalcogen\ atom$ [Example: S, Se, Te]

- Example: MoS_2, WS_2, $MoSe_2$ etc.

<u>**Structure**</u>

- One layer of M atoms is sandwiched between two layers of X atoms

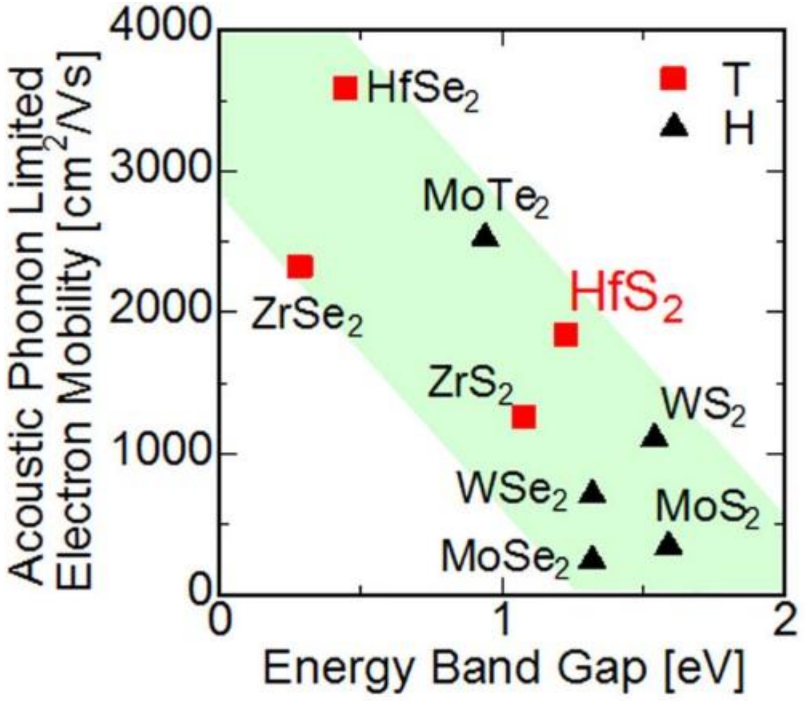
- Very thin 2D material [MoS_2 monolayer is only 6.5 Å thick]



Source: Wikipedia

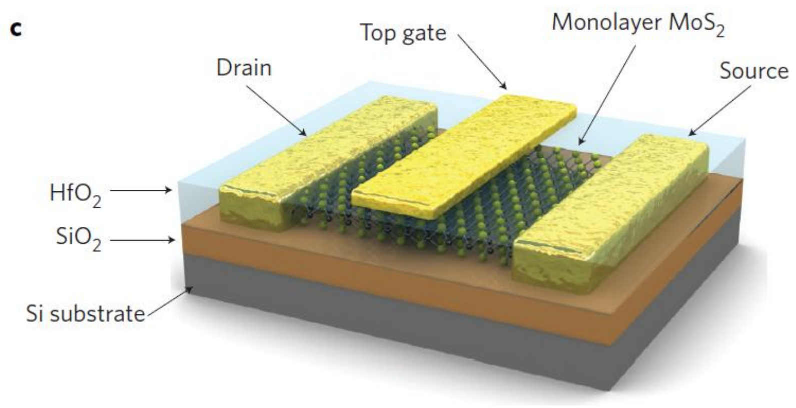
TMD: Properties

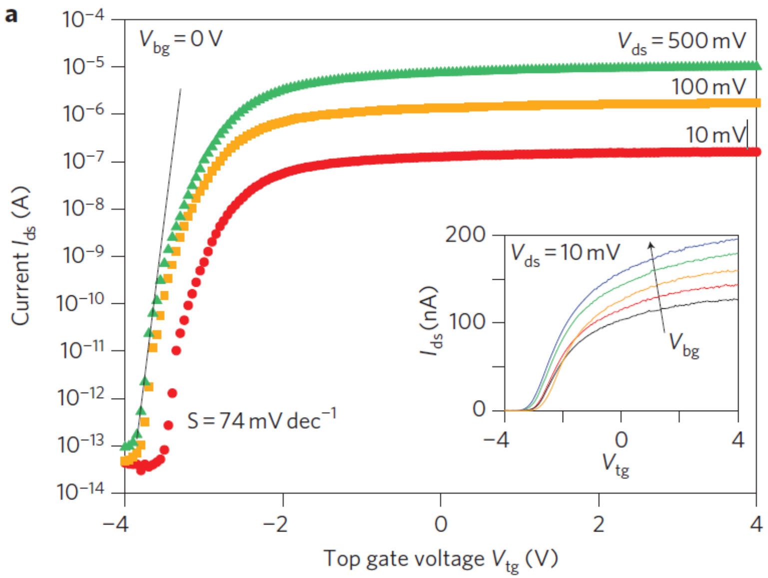
- Non-zero band gap
- Mobility comparable to silicon
- Larger effective mass



Source: Kanazawa, Toru, et al. "Few-layer HfS2 transistors." *Scientific reports* 6 (2016).

TMD: Transistors





- Low OFF-state current can be obtained
  - Non-zero bandgap
- High ON-current can be obtained
  - High DOS
  - Ballistic Conduction
- High ON-current/OFF-current ratio can be obtained
  - As high as 10^8 been reported

<u>Challenges</u>

- Fabrication at large scale with controlled properties

Source: Radisavljevic, Branimir, et al. "Single-layer MoS2 transistors." *Nature nanotechnology* 6.3 (2011): 147-150

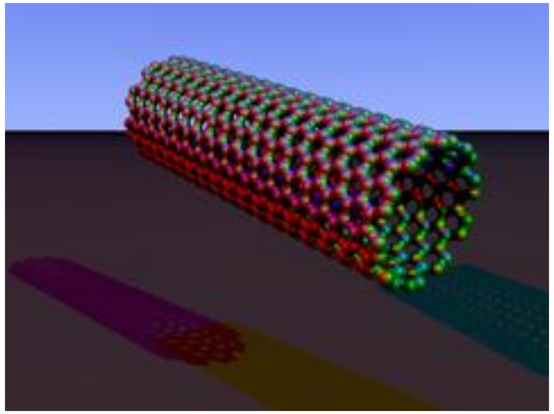
1-D Devices

CNT-based FETs

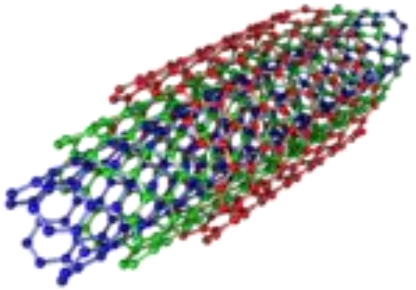
CNT: Basics

- S. Iijima first reported the formation of needle-like tubular carbon structures in 1991

- Carbon nanotubes (CNTs) are ***hollow cylindrical nanostructures of carbon*** in which the length is normally several orders of magnitude greater than the diameter



- Two types
  - ***Single-walled Carbon Nanotube*** (SWCNT): single shell of carbon atoms
  - ***Multi-walled Carbon Nanotube*** (MWCNT): multiple concentric shells

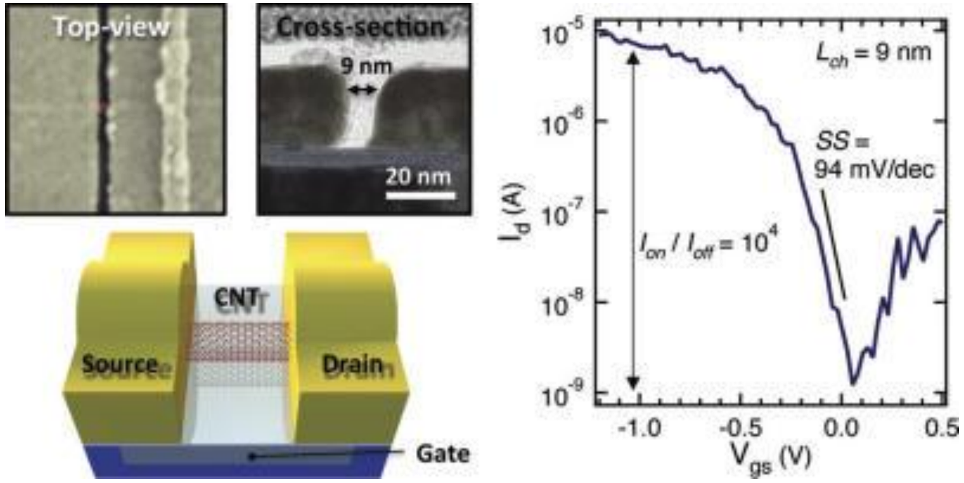


CNT: Properties

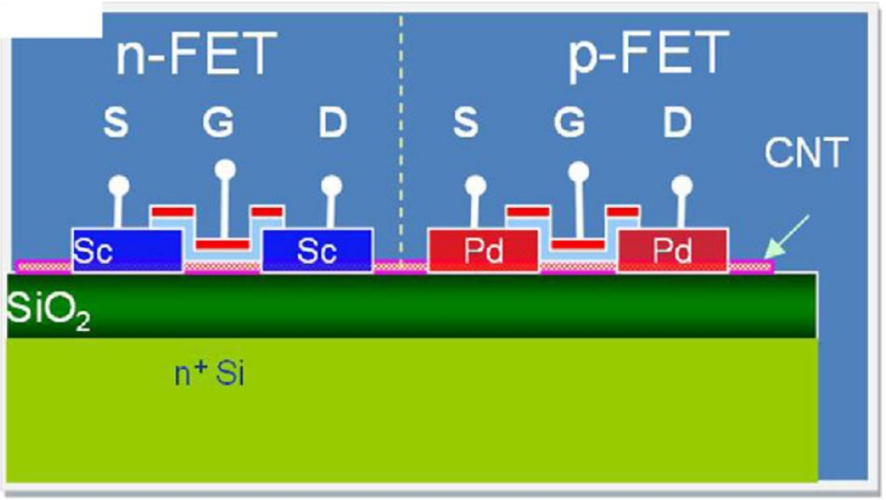
- Very strong material
- Can carry very high currents and inert nature: can be used as interconnects
- Good thermal conductivity
- Long mean free paths in carbon nanotubes
    - Ballistic current possible
- High mobility

CNT: Transistors

- Bandgap of CNT allows obtaining good transistor characteristics at smaller channel lengths
  - ➢ High mobility
  - ➢ Ballistic conduction



- Chemical doping cannot be used
- Metal/CNT junction properties has a major role in determining the characteristics of device



Source: L-M. Peng, Z. Zhang and S. Wang, "Carbon nanotube electronics: recent advances", *Materials Today*, vol-17, pp. 433-442, Nov. 2014

CNT: Prospects

- Potential: can outperform silicon by improving the energy-delay product by more than an order of magnitude
- Computer built entirely using CNT-based transistors has been demonstrated [2013]
- Scaling of CNT to 5 nm possible

- Challenge: Non-perfect control of CNT growth
  - As-grown CNTs usually contain both semiconducting and metallic CNTs, which are not suitable for electronics applications.
  - Solutions:
    - ✓ Selective growth of semiconducting-enriched CNTs
    - ✓ Controlled removal of metallic CNTs
  - High purity (99%) semiconducting CNTs have been obtained

IRDS

- IRDS: International Roadmap for Devices and Systems

- Initially, ITRS (or International Technology Roadmap for Semiconductors) was produced annually
  - Serve as the main reference into the future for university, consortia, and industry researchers to stimulate innovation in various areas of technology
- In May 2016, the International Roadmap for Devices and Systems (IRDS™

- Goals
  - Create Roadmap: Identify key trends related to devices, systems, and all related technologies
  - Innovation: Determine generic devices' and systems' needs, challenges, potential solutions, and opportunities for innovation
  - Collaboration: Encourage related activities worldwide through collaborative events

Summing Up

Good Luck for

All your future Endeavors!

Excel in whatever you choose to do!